Microscopic model of avalanches and oscillations in neural networks

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Outline

1. **Intro:** Avalanches in general and in neuronal networks
2. **Model:** Ising model with negative feedback
3. **Data** (MEG of human brain) vs. model
4. **Conclusion**

Goal: Develop and study a <u>minimal</u> physics model of neuronal networks in which <u>oscillations and avalanches</u> coexist, and which is analytically tractable.

Avalanches in reality

- On 29.12.2020. a 6.4 magnitude earthquake hit Petrinja, Croatia.
- 6 dead, 26 injured, at least half of the city destroyed.

- **Intuition:** weaker earthquakes are more frequent than stronger ones
- **Data?**

me

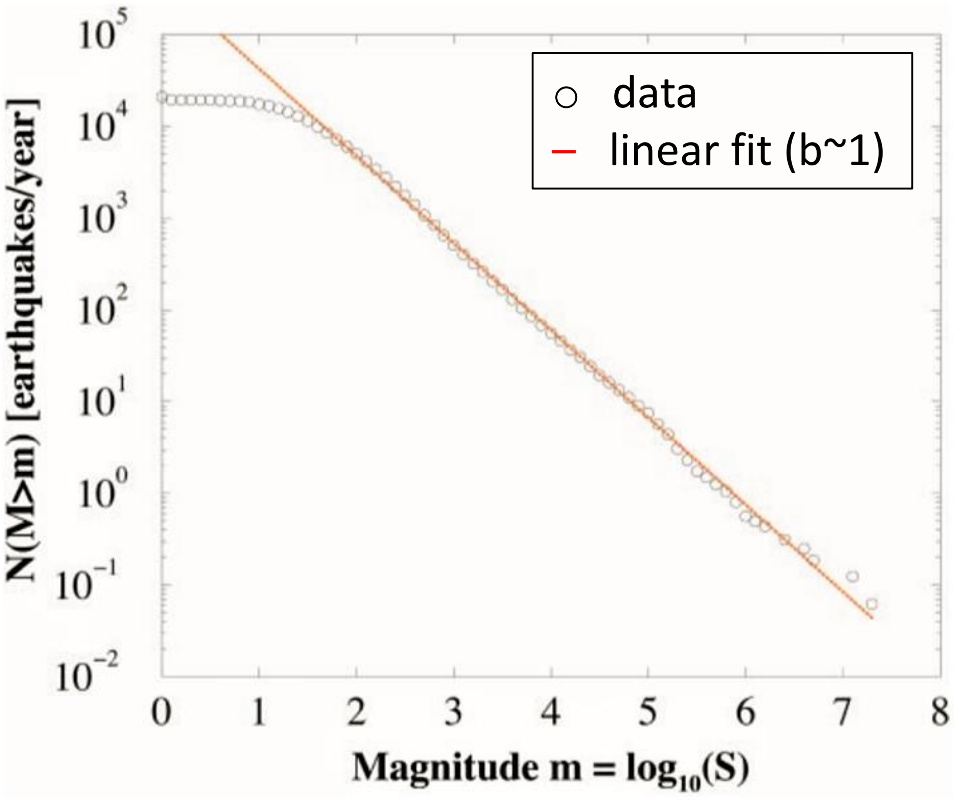
The Sun, „Croatia earthquake map: Where was the epicenter and how many people have died?“, retrieved 07.01.2021.

Avalanches in reality

- On 29.12.2020. a 6.4 magnitude earthquake hit Petrinja, Croatia.
- 6 dead, 26 injured, at least half of the city destroyed.

- **Intuition:** weaker earthquakes are more frequent than stronger ones
- **Data:**
  Gutenberg–Richter law - magnitude distribution follows a **power-law**

Gutenberg–Richter law



Christensen et al. Unified scaling law for earthquakes, PNAS, 2002, vol. 99, 2509-2513

Avalanches in general

- **Present in excitable, coupled systems:**
  sandpiles, earthquakes, fires, nuclear reactions, neural networks...

- **Typical dynamics:**
  - a unit sums excitation from nearby units...
  - if above threshold, unit „fires“ and spreads excitation...
  - (possibly) leading to a cascade of „firing“ – an avalanche

Avalanches in general

- Spatial and temporal distributions of avalanches usually well described by **power-laws.**

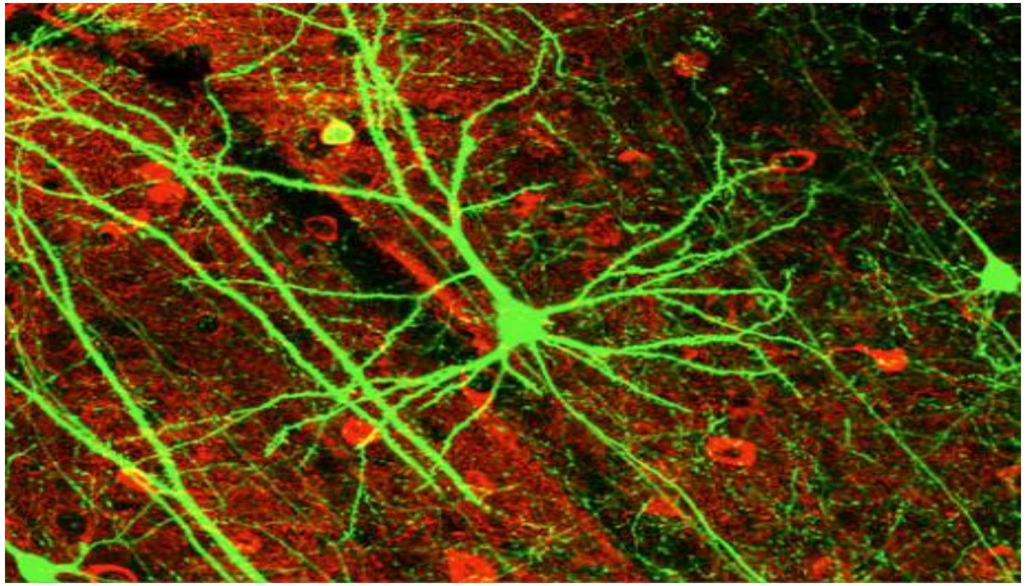
- Indicative of a system in a **critical state** and **scale-free** dynamics [Bak, Tang, Wiesenfeld]

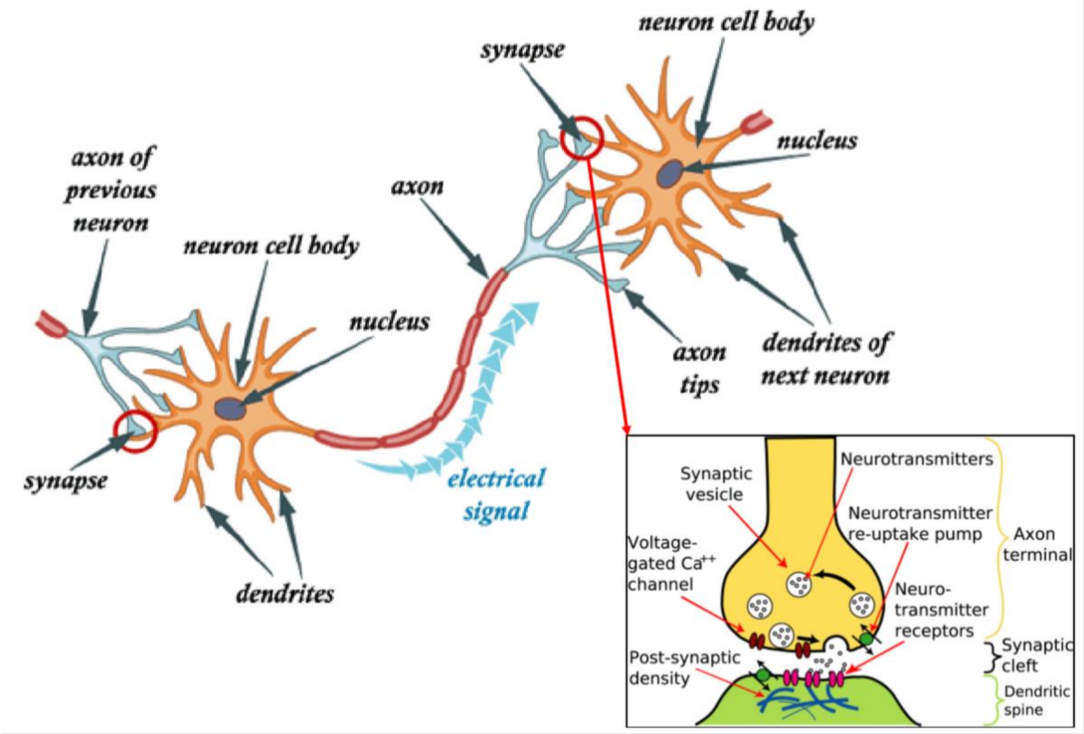
Bak, Tang, Wiesenfeld, Self-Organized Criticality: An Explanation of 1/f Noise, PRL 1987

Hesse, Gross, Self-organized criticality as a fundamental property of neural systems, Front. Syst. Neurosci., 2014

Neurons and networks

- Neurons are electrical excitable cells and the basis of the nervous system.
- Neuron **„sums“** the excitation from neighboring neurons (connected through synapses), and if above **threshold**, it is excited (**spikes**) and **transmits** the excitation to neurons „downstream“.
- Neuronal networks are big: brain has ~100 bilion neurons, each with ~1000 synapses.

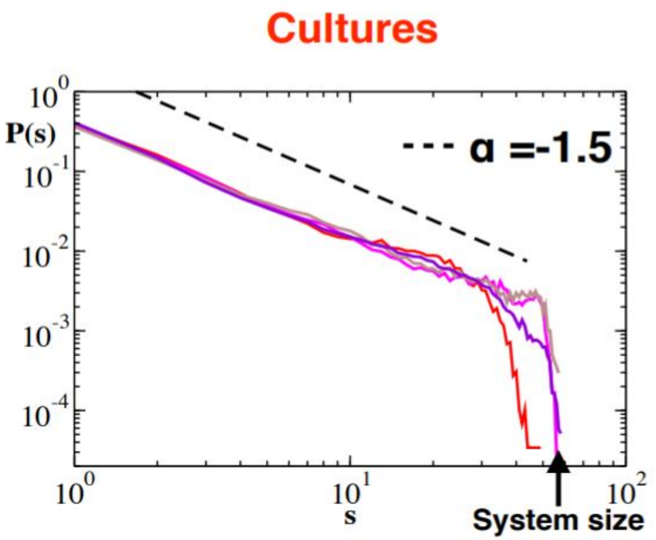


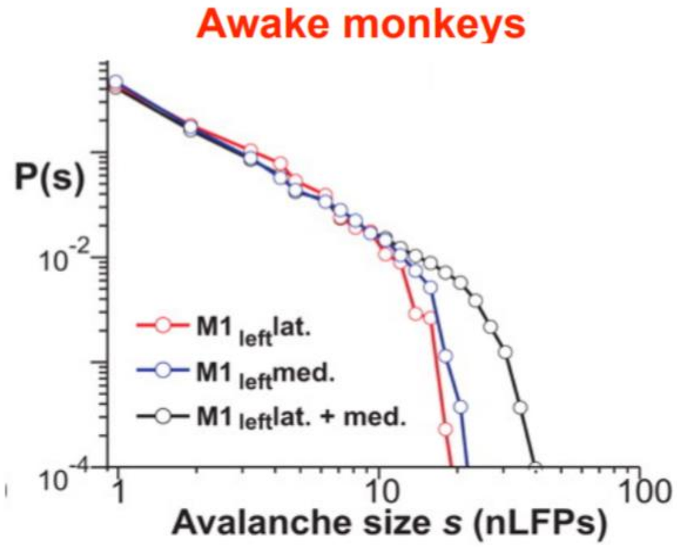


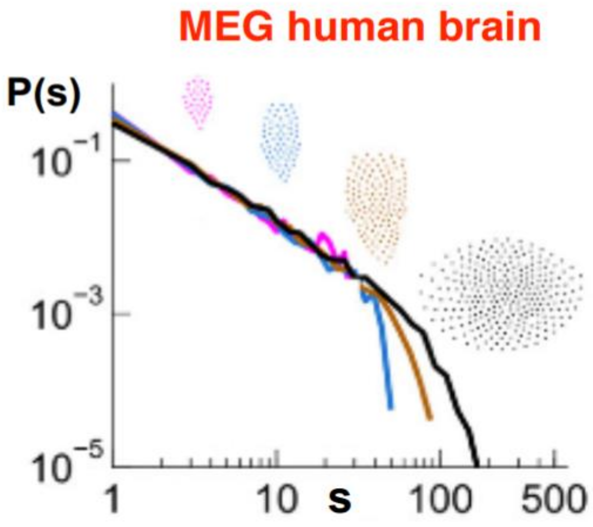
Lee WCA et al, Dynamic Remodeling of Dendritic Arbors in GABAergic Interneurons of Adult Visual Cortex. PLOS Biology, 2005

Avalanches in neuronal networks

- **Long-standing hypothesis:**
  neuronal networks (brain) operate at a critical point
- Seemingly supported by experiments [Beggs, Plenz]





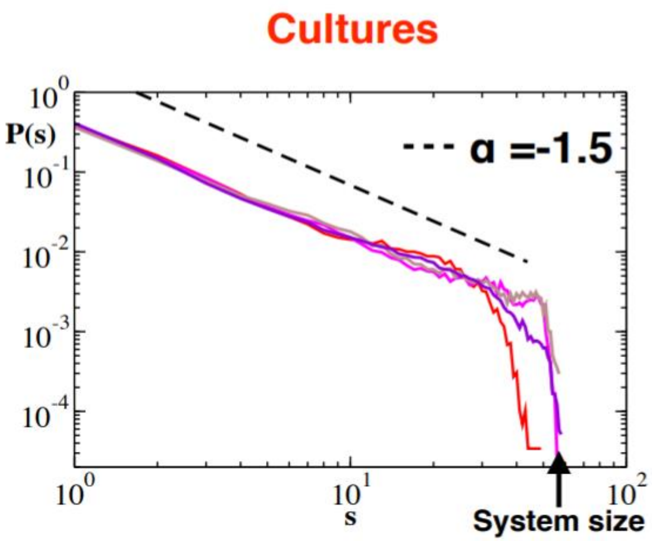


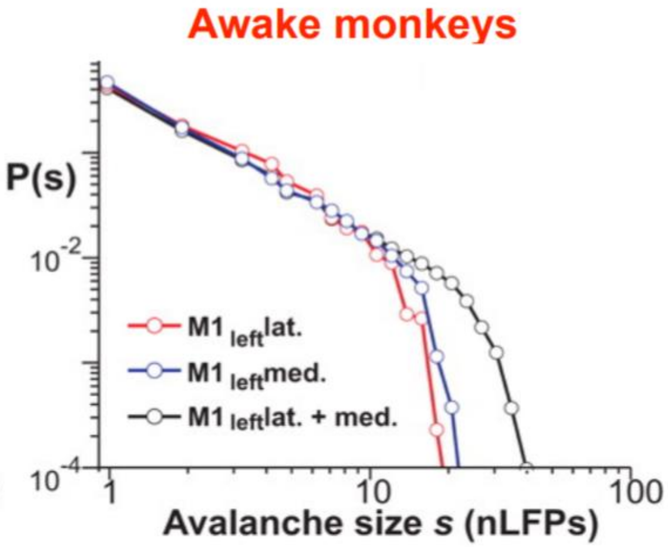
Beggs, Plenz, Neuronal Avalanches in Neocortical Circuits. The Journal of Neuroscience, 2003

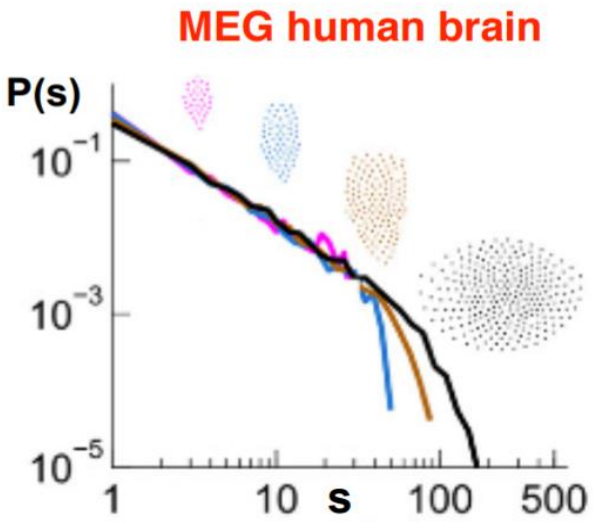
Wilting, Priesmann, 25 of criticality in neuroscience, Current Opinion in Neurobiology 2019

Avalanches in neuronal networks

- **Questions:**
  Does criticality have a function? How is it reached?
  Is the activity critical or near critical?
  Relation with other modes of brain activity?







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Avalanches in neuronal networks

- **Questions:**
  Does criticality have a function? How is it reached?
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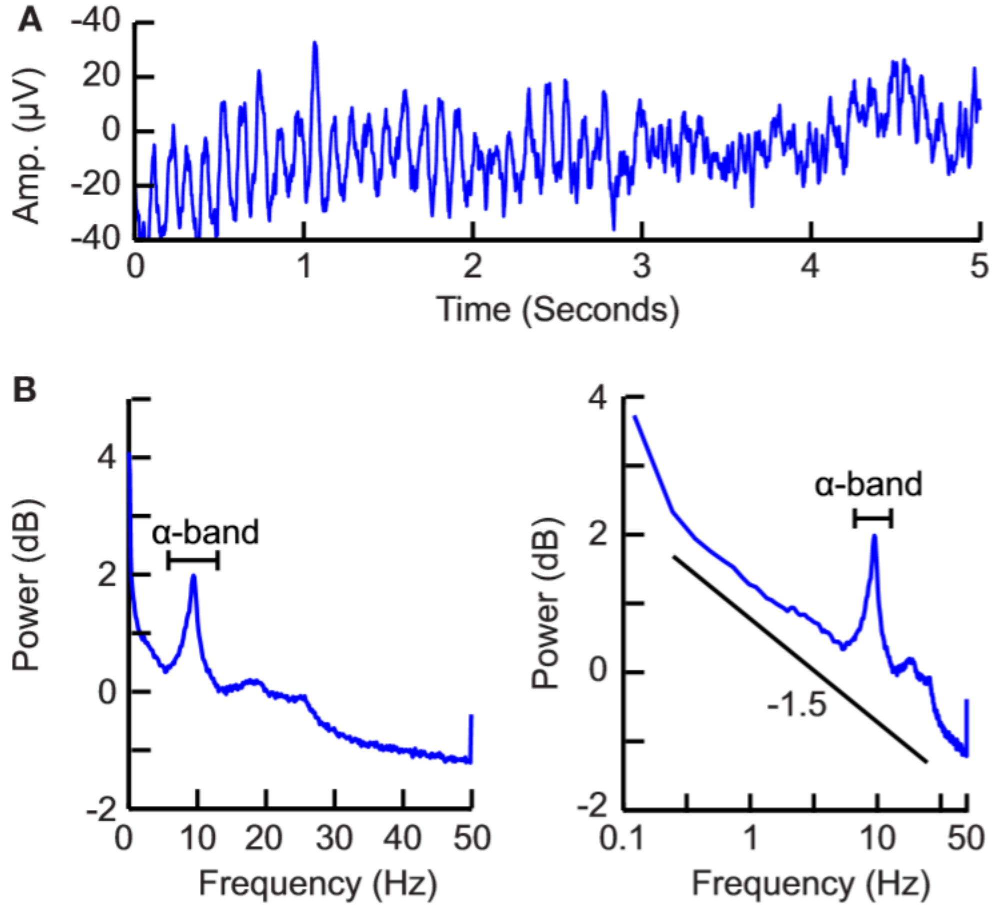
  Relation with other modes of brain activity?

- Possible benefits (and perils):
  - Long-range correlations (needed for coordinated coherent activity)
  - Large variability in achievable activity patterns (information representation)
    ... but this leads to critical slowing-down (slow „computation“)
  - Maximization of information transmissions
    ... but this leads to runaway excitations as in epilepsy

Oscillations in neuronal networks

- **Oscillations** coexist with avalanches in the brain!

- **Questions:**
  What is their function?
  What controls their dynamics?
  Relation with other modes of brain activity?



Hardstone et al. Detrended fluctuation analysis: a scale-free view on neuronal oscillations, Front. Physiol., 2012

Goal

- Develop and study a minimal physics model of neuronal networks in which oscillations and avalanches coexist, and which is analytically tractable.

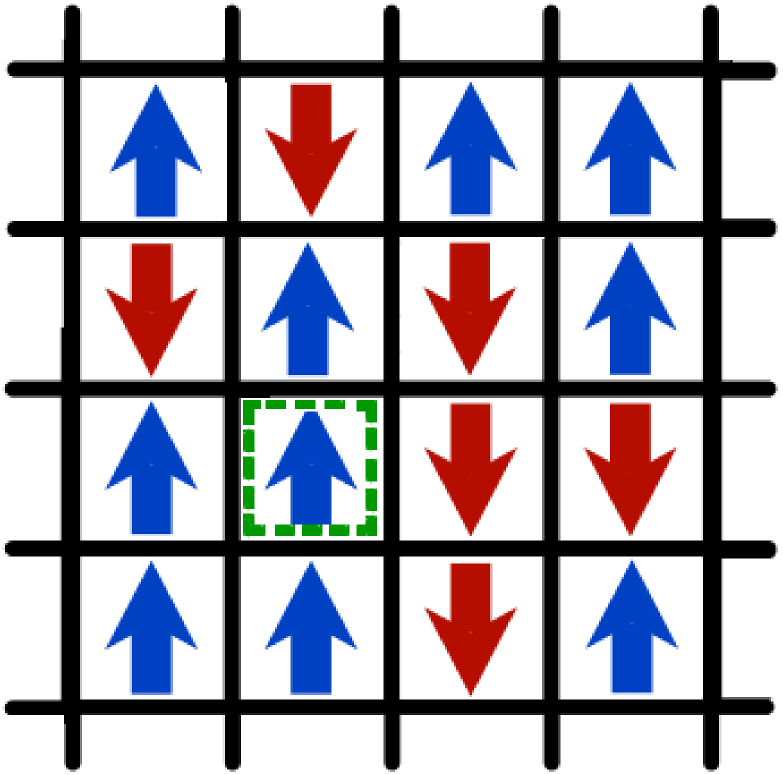
- Answer some of previous questions and better understand how the brain function.

- Possibly use this knowledge in a practical way (e.g. deviation from the „optimal“ power-law exponent could suggest a pathology)

Our model: Ising with a negative feedback

- Various model exist based on how stochastic vs. deterministic and mean-field (averaged) vs. spiking (local) they are.

- We used a **2D Ising model** (stochastic) **with a global negative feedback** (deterministic) applied by the external magnetic field.



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Ising model (classical)

- Energy of a single spin:

$$E_{ij} = -\boldsymbol{B}\sigma_{ij} - \sum_{mn=<ij>} \boldsymbol{J}\sigma_{ij}\,\sigma_{mn}$$

magnetic field
external drive

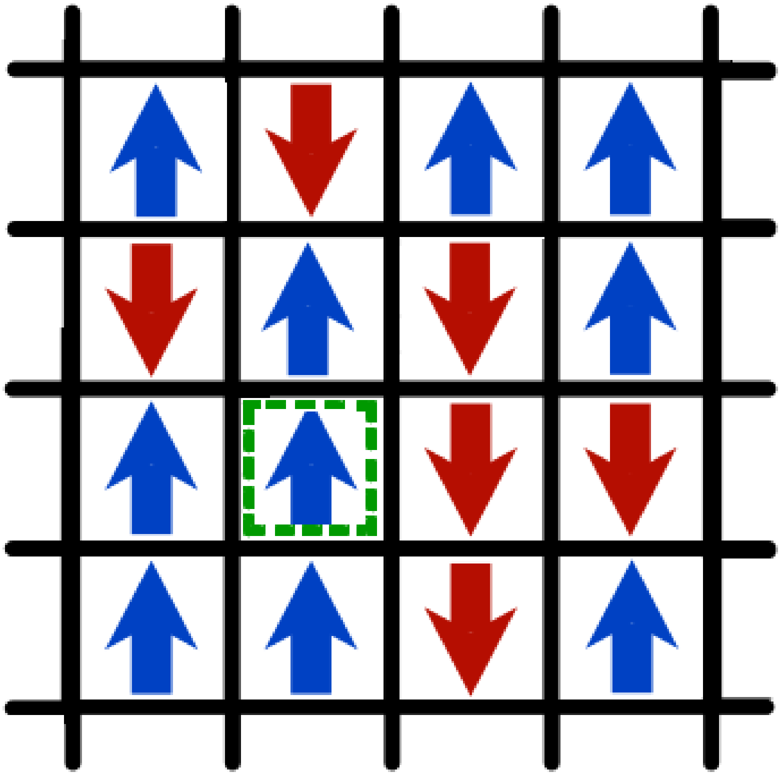
coupling strength
neuron connectivity

- Total magnetization: $m = \sum \sigma_{ij} / N$

spin value
state of neuron

- Flip probability: $p_{S_1 \rightarrow S_2} = e^{-\beta \Delta E_{12}}$

Inverse temperature



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Ising model (classical, B=0)

- System state depends on the relative strength of order (spin-spin coupling $\boldsymbol{J}$) versus thermal disorder ($\boldsymbol{\beta}$).
- Results in either **paramagnetic** or **ferromagnetic** phase.
- Phase transition at the critical temperature β_c.

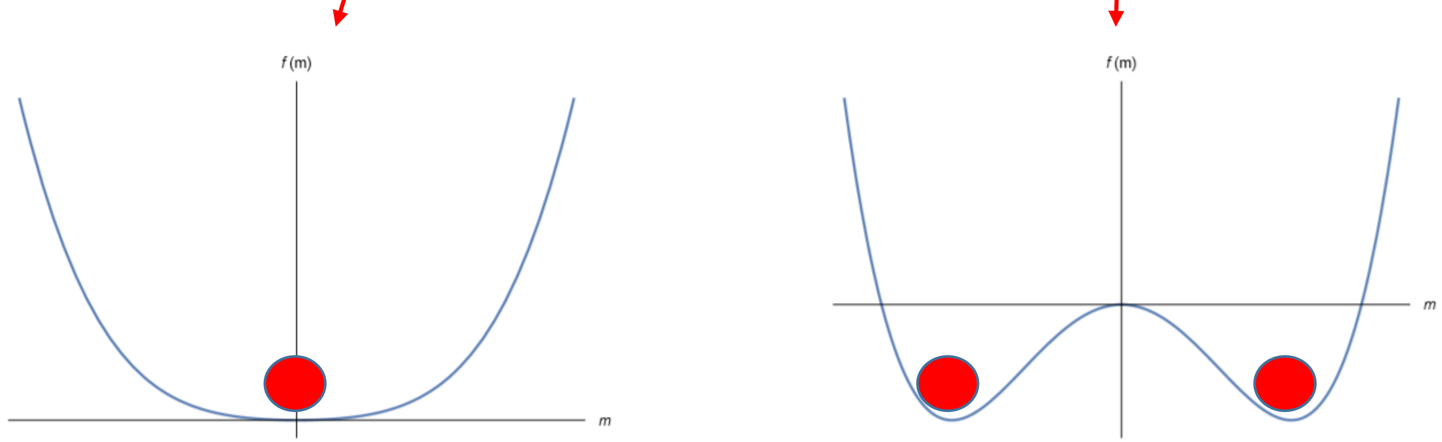
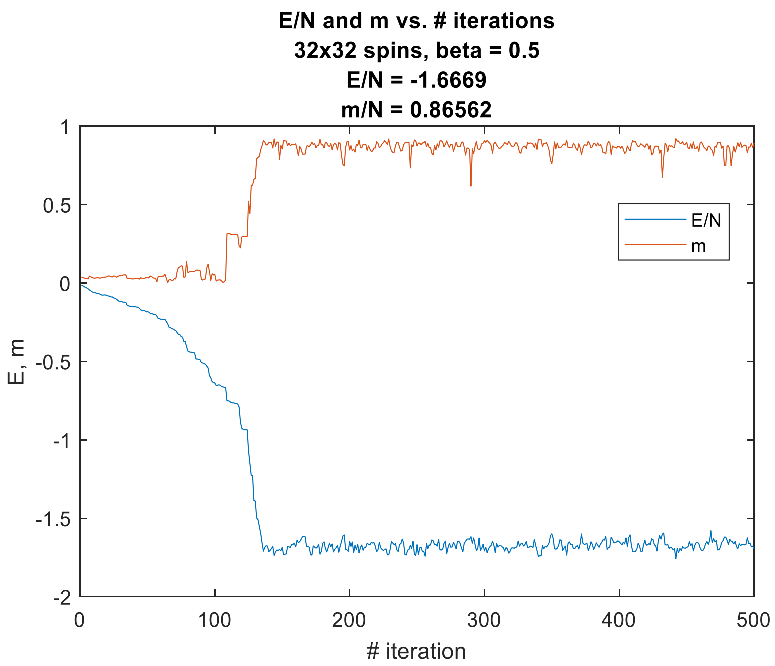


Figure 1: Free energy when $T > T_c$

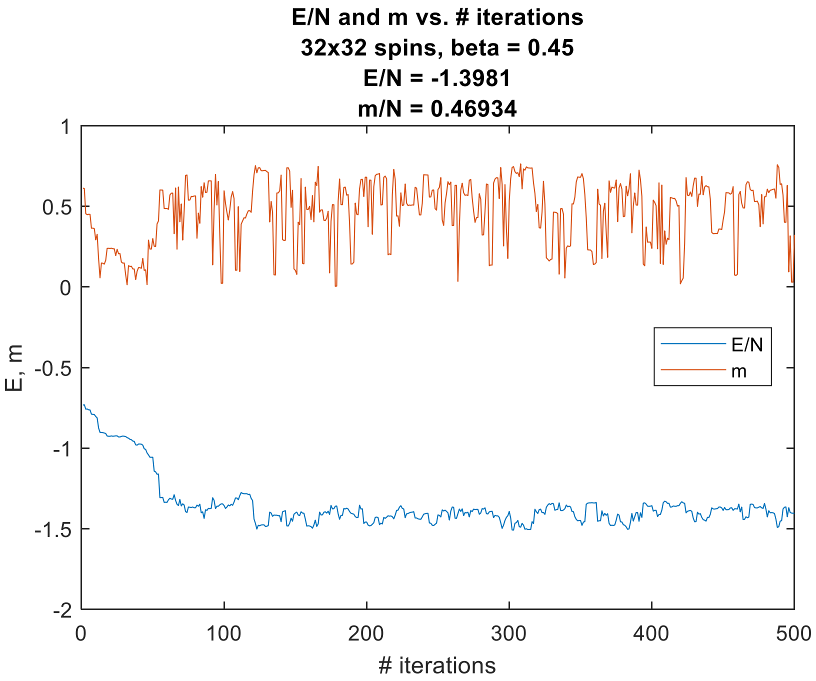
Figure 2: Free energy when $T < T_c$

David Tong, Statistical Field Theory course lecture notes, 2017

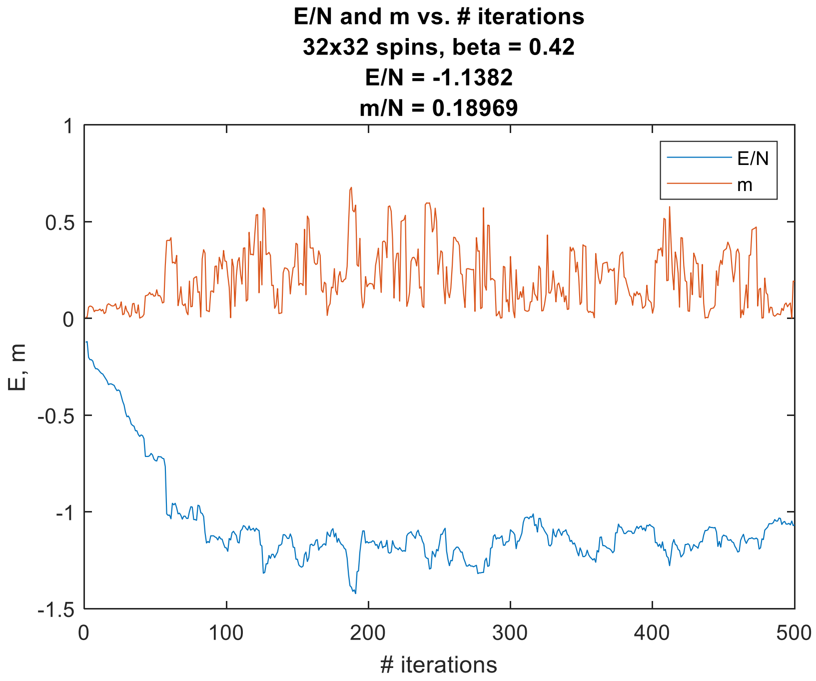
Ising model (classical)
Monte Carlo simulation, Metropolis algorithm, B=0



$\boldsymbol{\beta > \beta_c}$



$\boldsymbol{\beta \approx \beta_c}$



$\boldsymbol{\beta < \beta_c}$

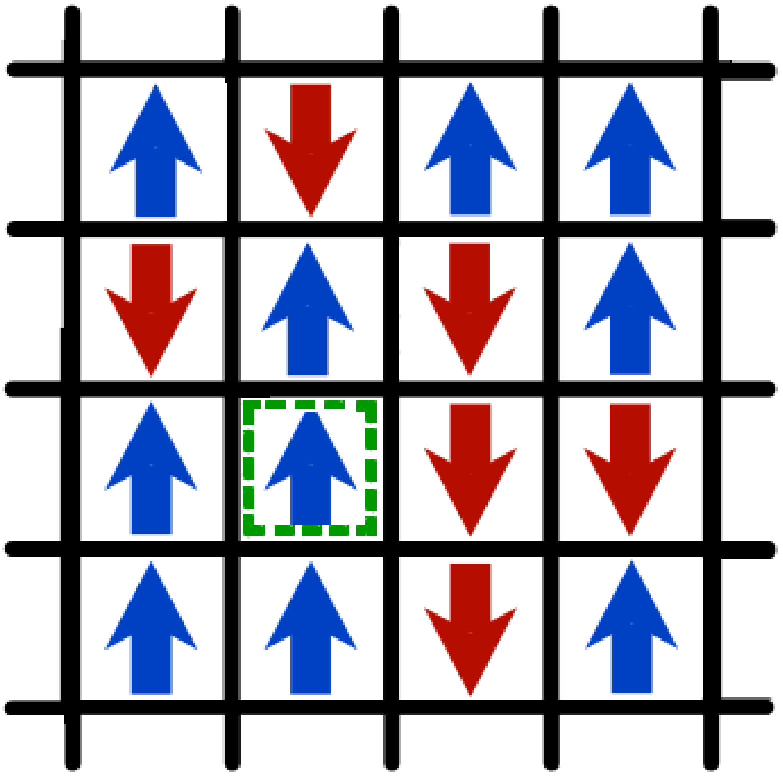
Ising model (negative feedback)

- Negative feedback (neural inhibition) is applied by the external magnetic field (computed globaly, applied to every spin):

$$\frac{dB}{dt} = -\frac{m}{\tau}$$

feedback timescale

- **Intuition:**
  - for low temp, magnetic field grows in opposition to the magnetization, flips it at some point, and then the reverse happens,
  - for high temp, thermal fluctuations dominate the dynamics

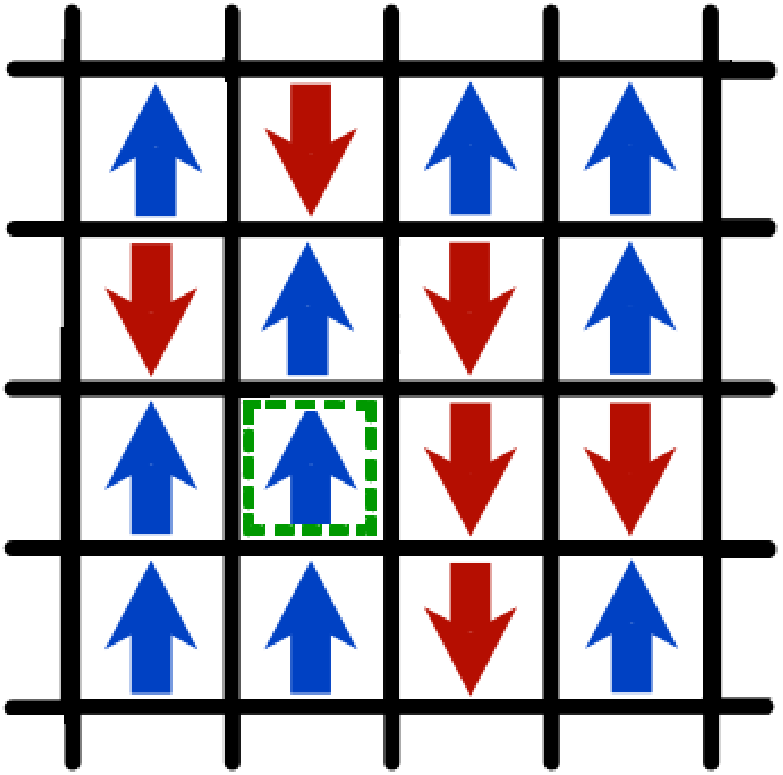


Ising model (negative feedback)

- Governing equation can be obtained for mean-field behavior (assuming linear response):

$$\ddot{\boldsymbol{m}} - \left(\beta_{rel} - 1 - \boldsymbol{m}^2\right)\dot{\boldsymbol{m}} + \frac{1}{\tau}\boldsymbol{m} = 0$$

- Van der Pol oscillator with Andronov-Hopf bifurcation (birth/death of periodic solution) at the phase transition ($\beta_{rel} = 1$)



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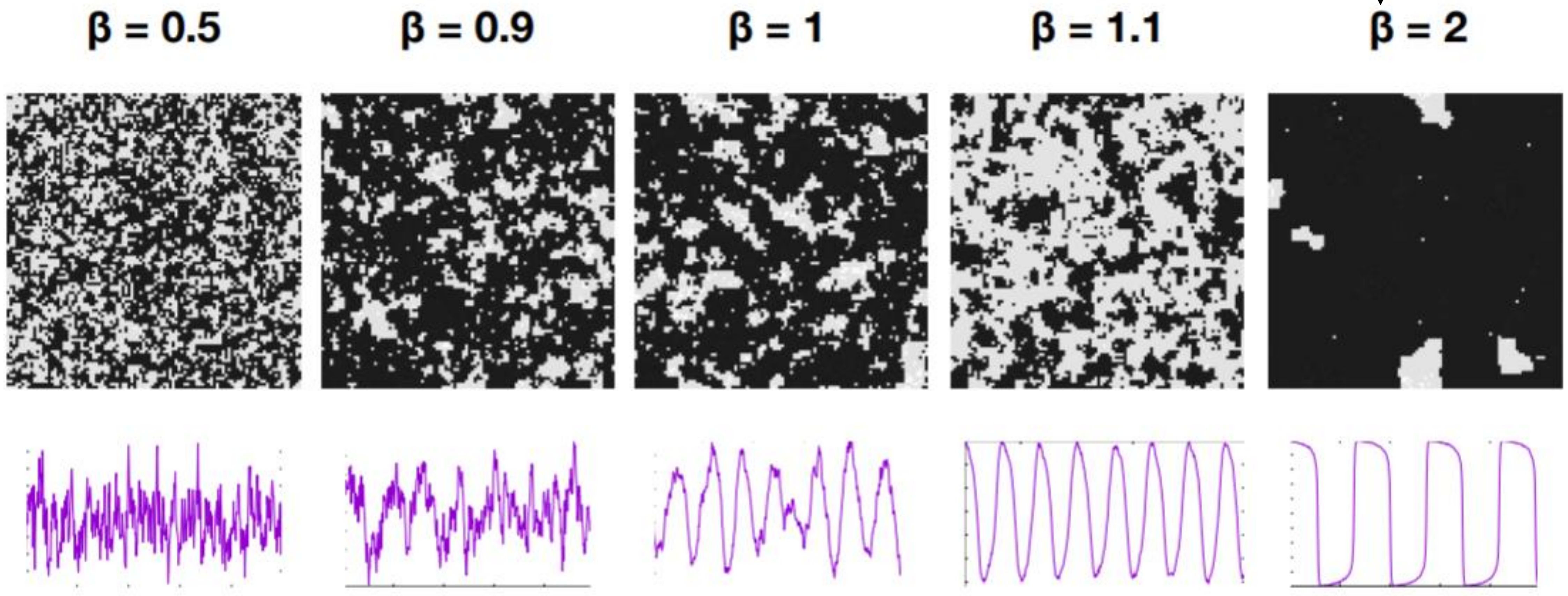
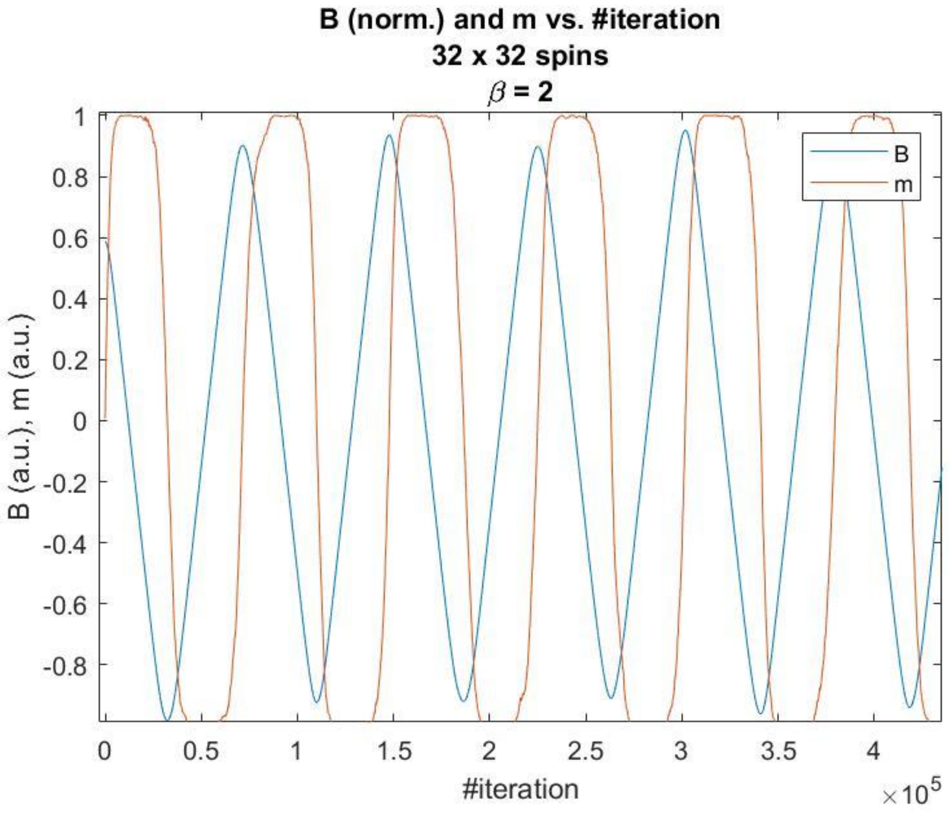
Monte Carlo simulations

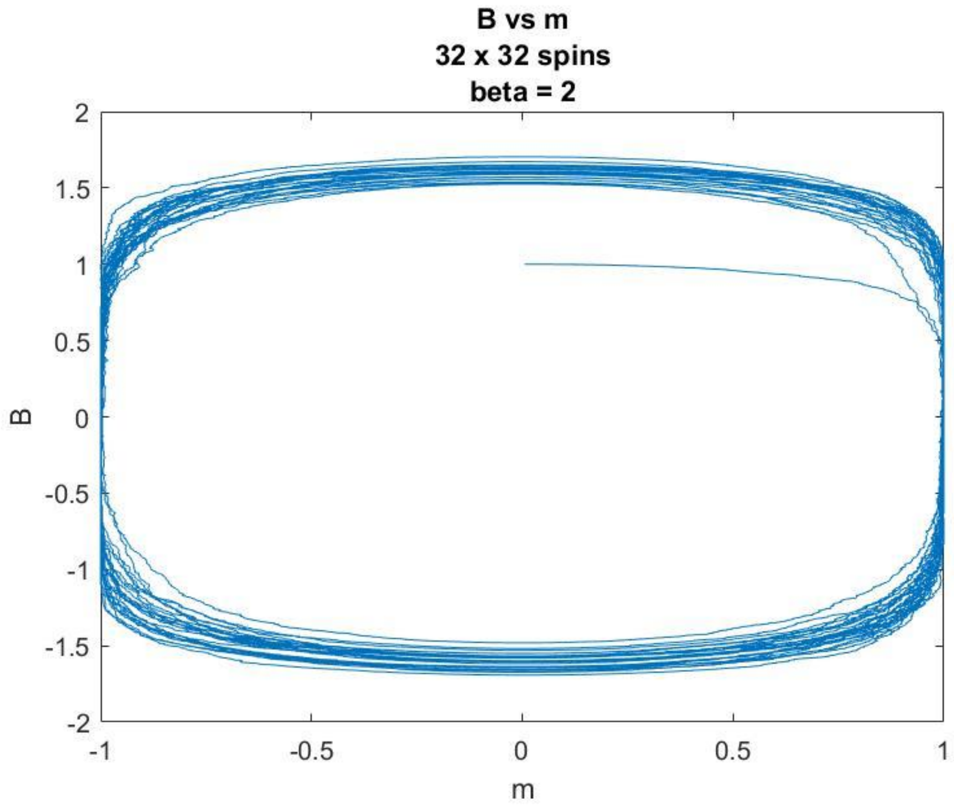
relative inverse temperature



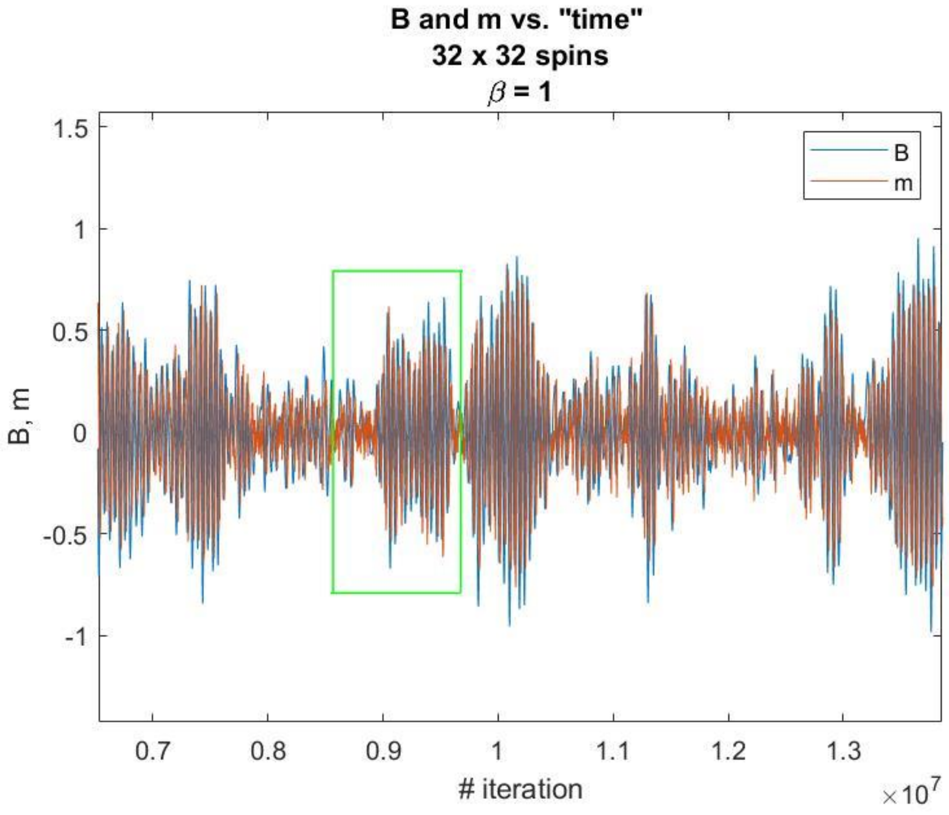
critical transition

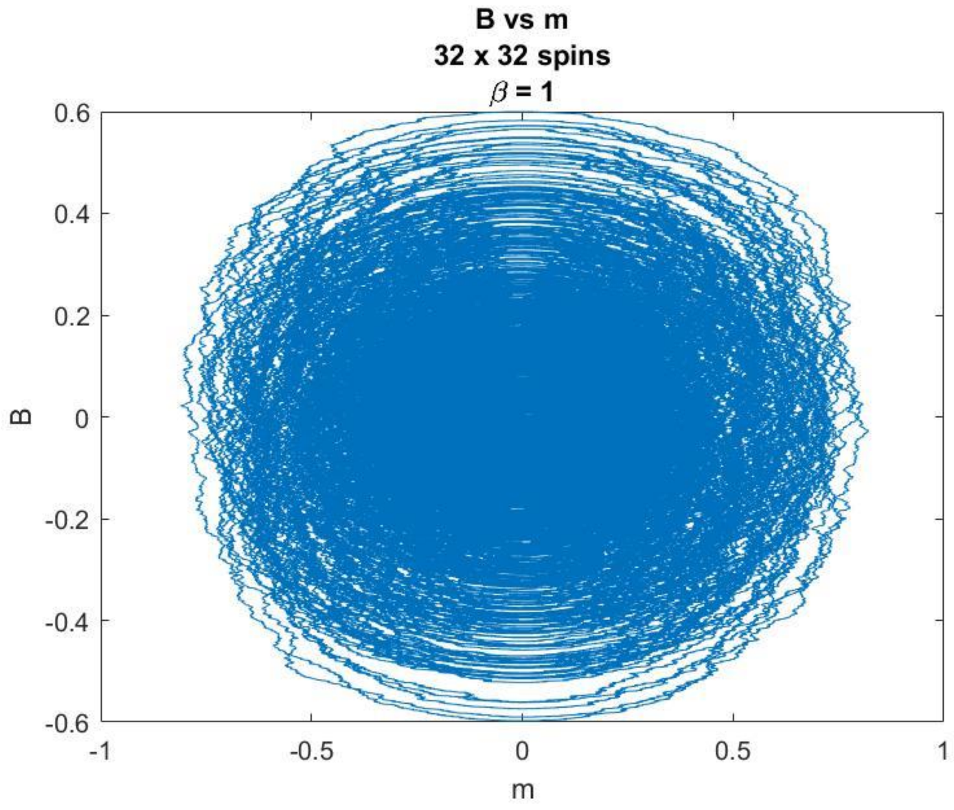
Monte Carlo simulations – low temperature



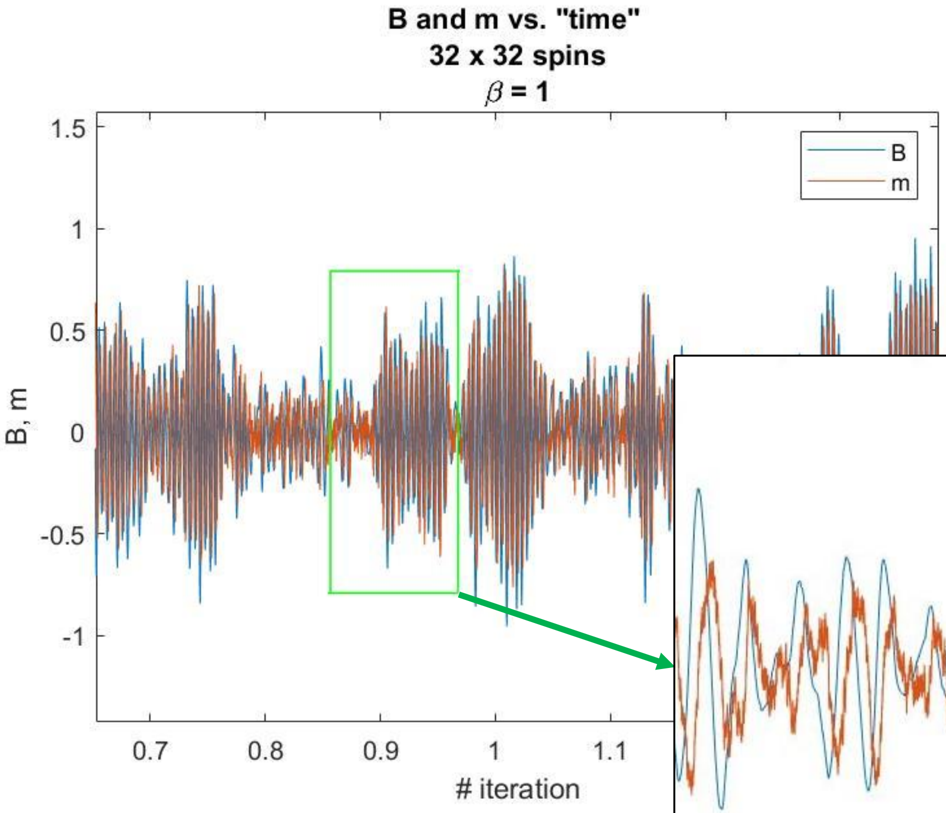


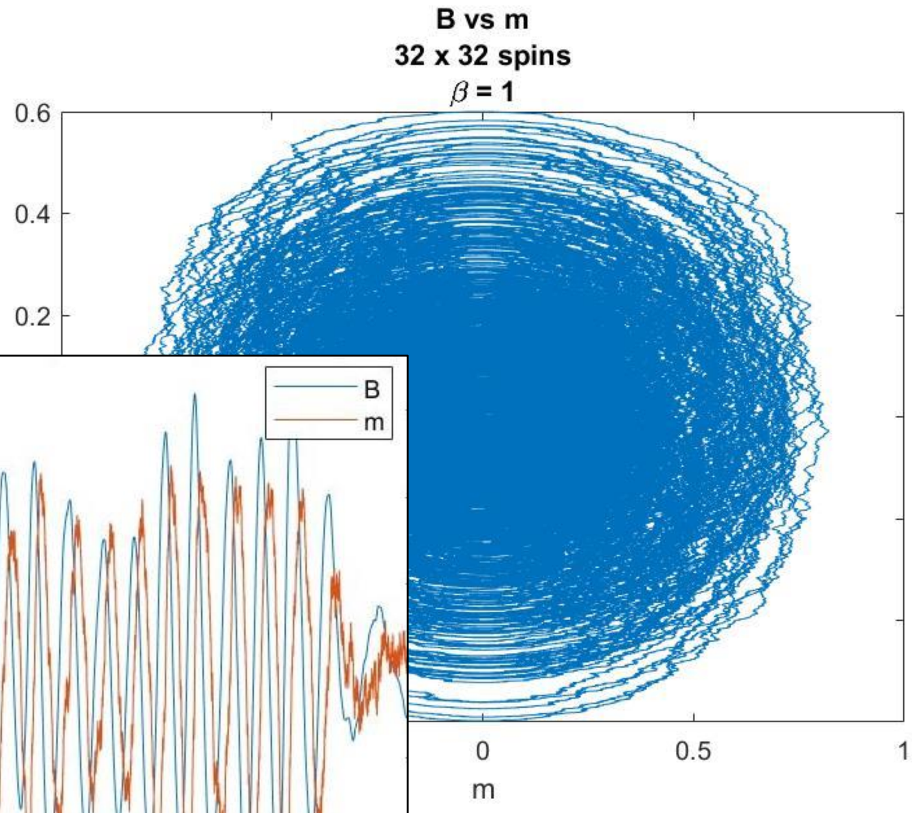
Monte Carlo simulations – critical temperature

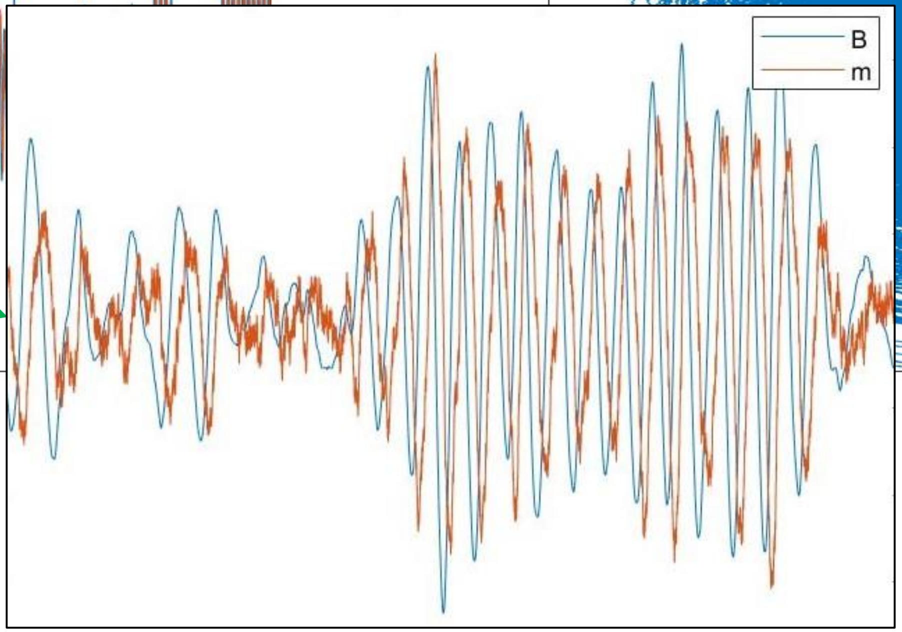




Monte Carlo simulations – critical temperature



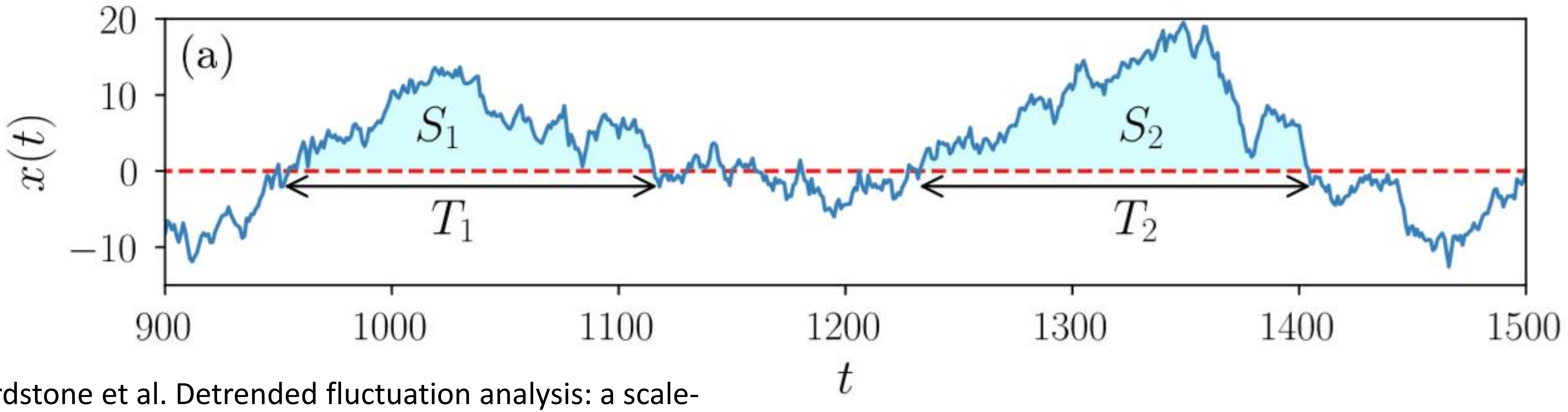




Oscillations + avalanches?

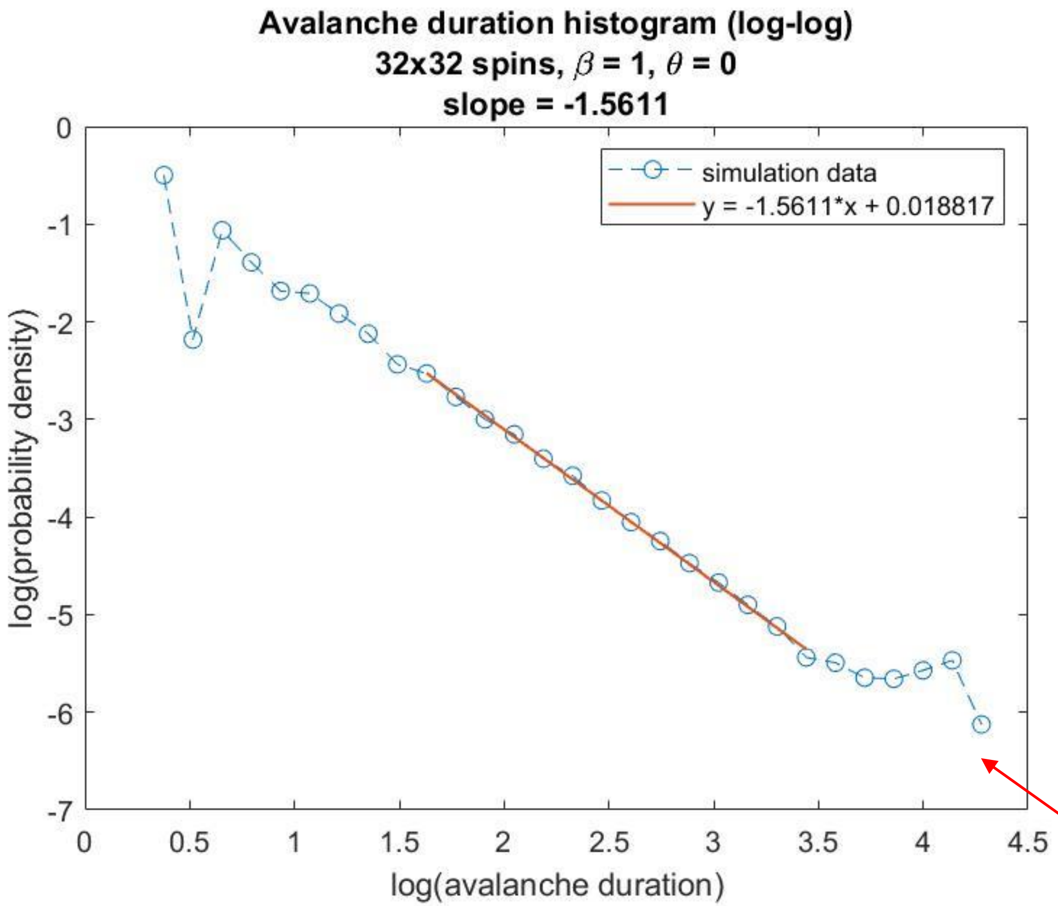
Avalanche definition

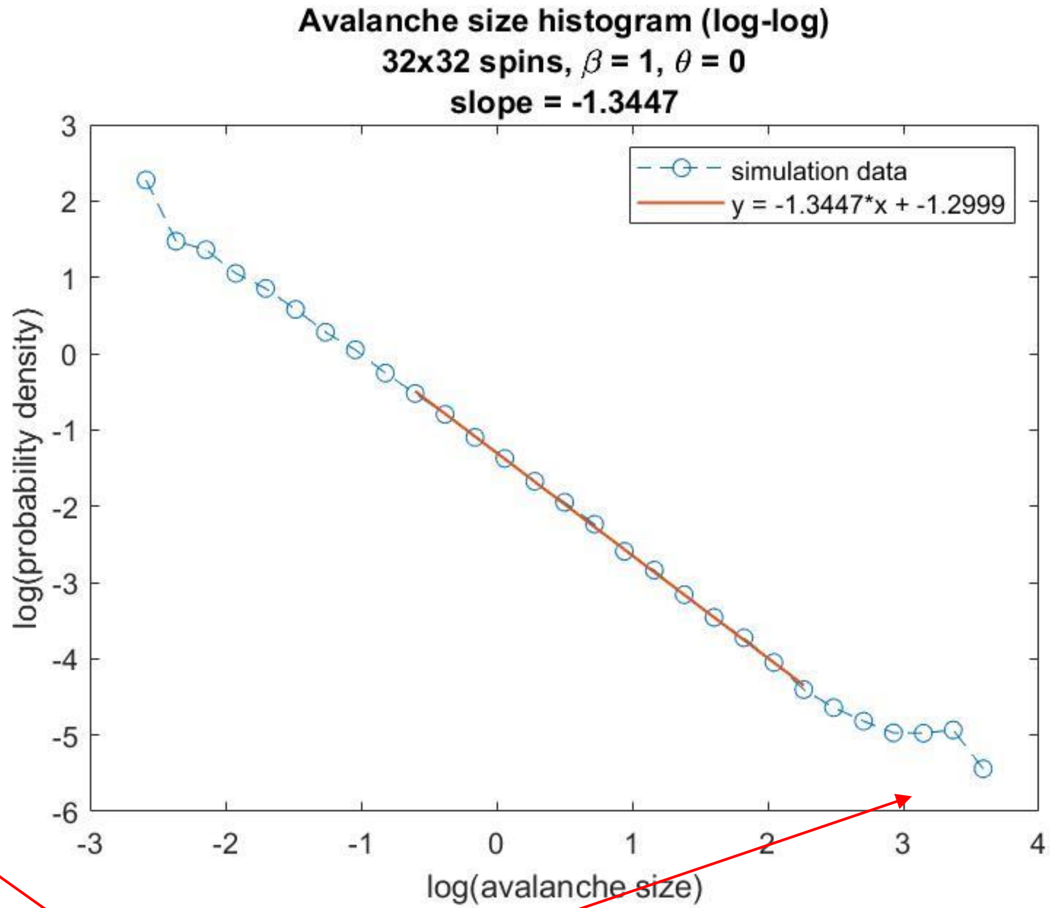
- Avalanche = **threshold crossings** of magnetization time series
- Threshold is a freely chosen parameter
- Size = area under the curve



Hardstone et al. Detrended fluctuation analysis: a scale-free view on neuronal oscillations, Front. Physiol., 2012

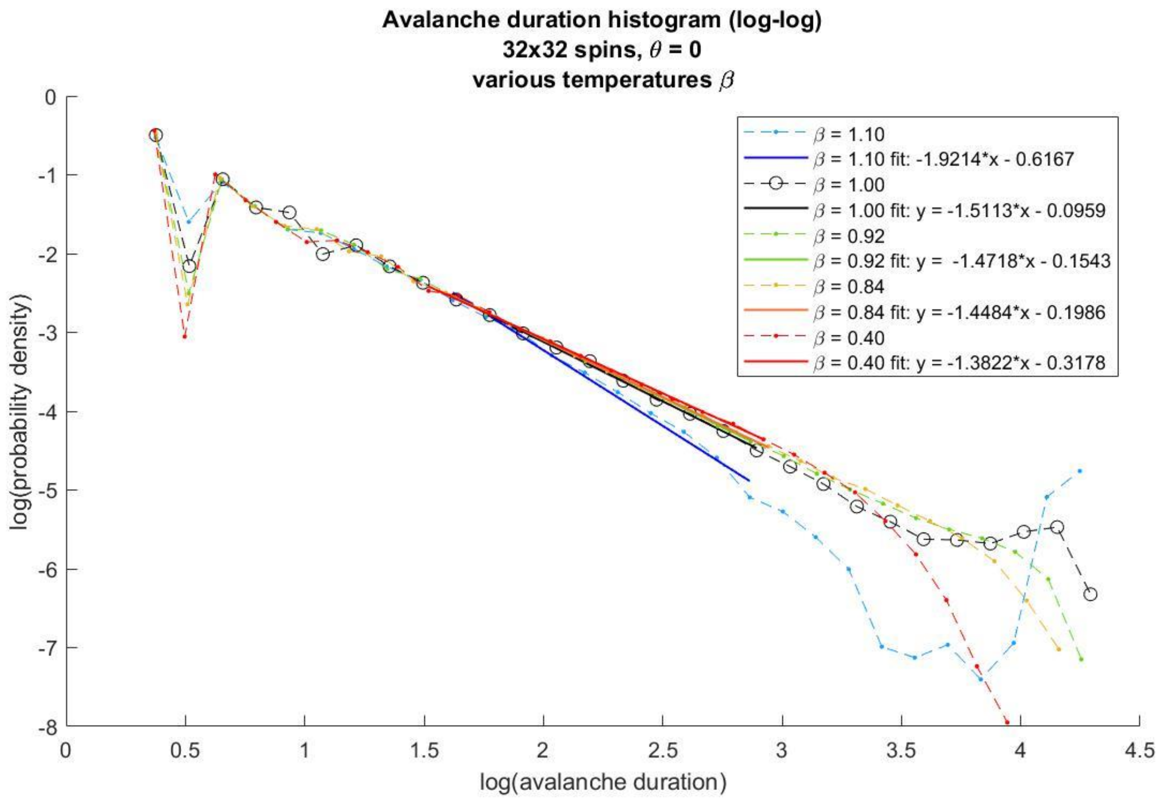
Avalanche histograms (duration, size)

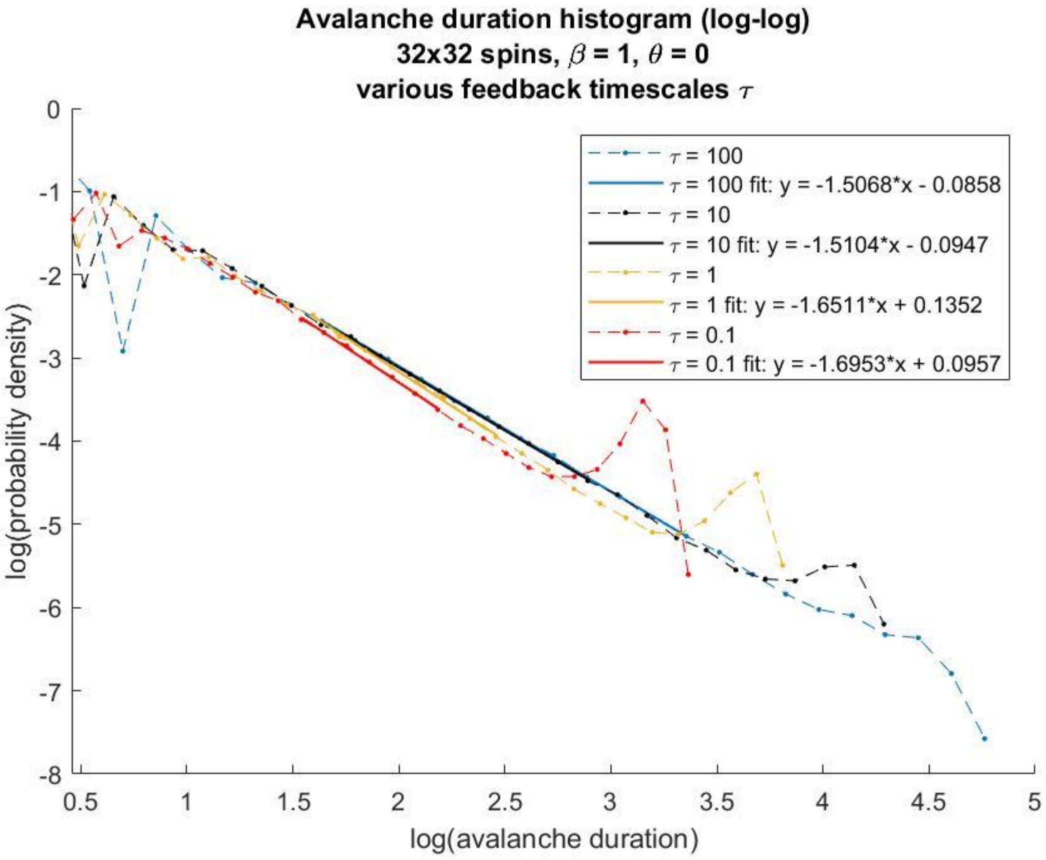




oscillations

Avalanche histograms vs. **temperature** and **feedback timescale**





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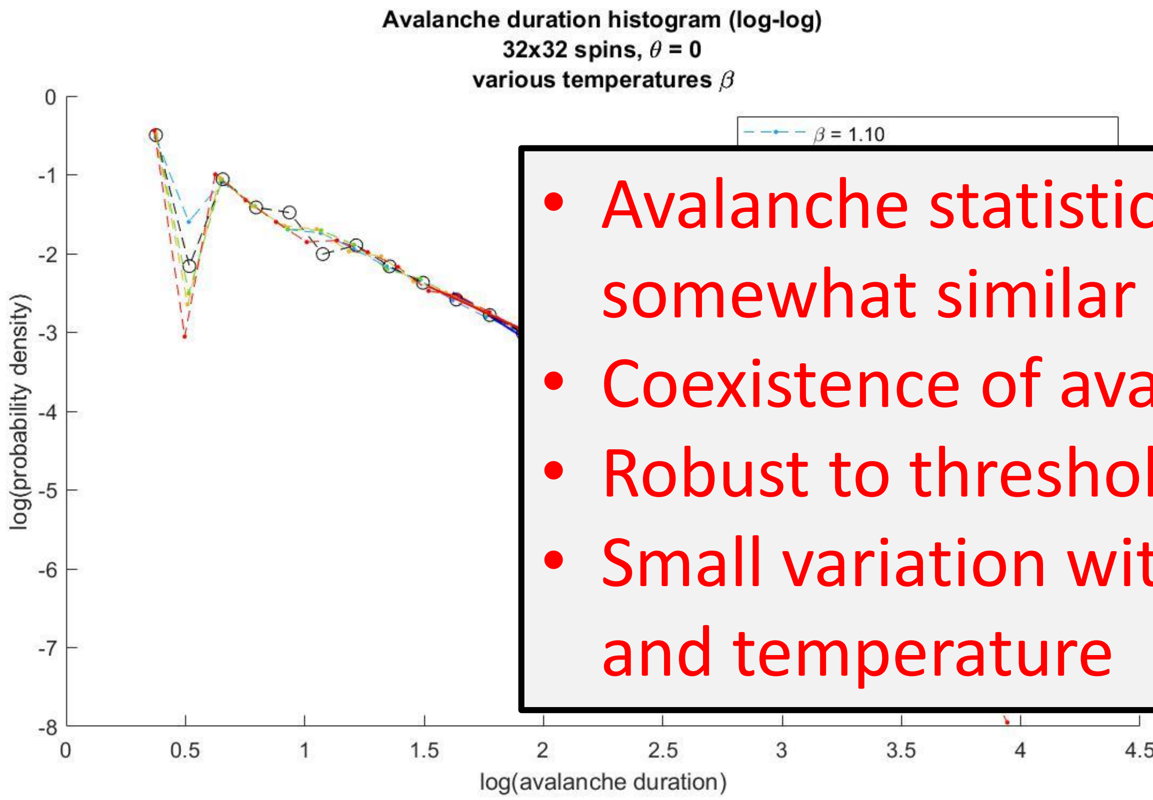
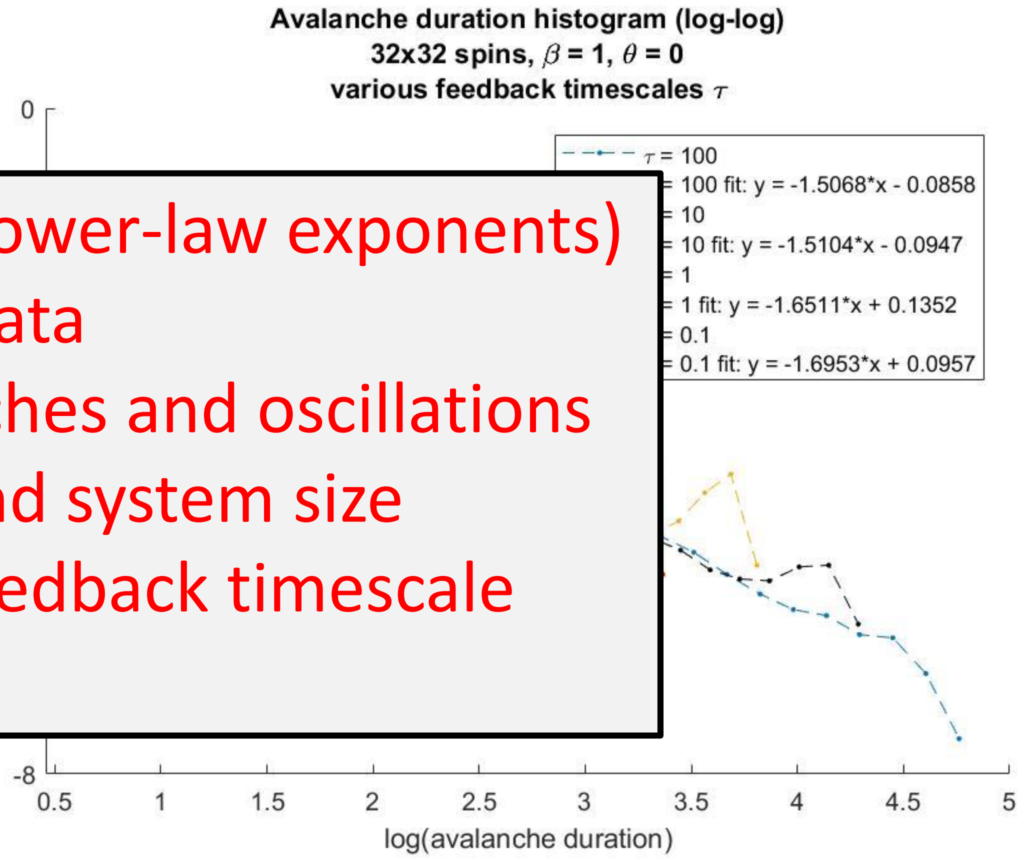


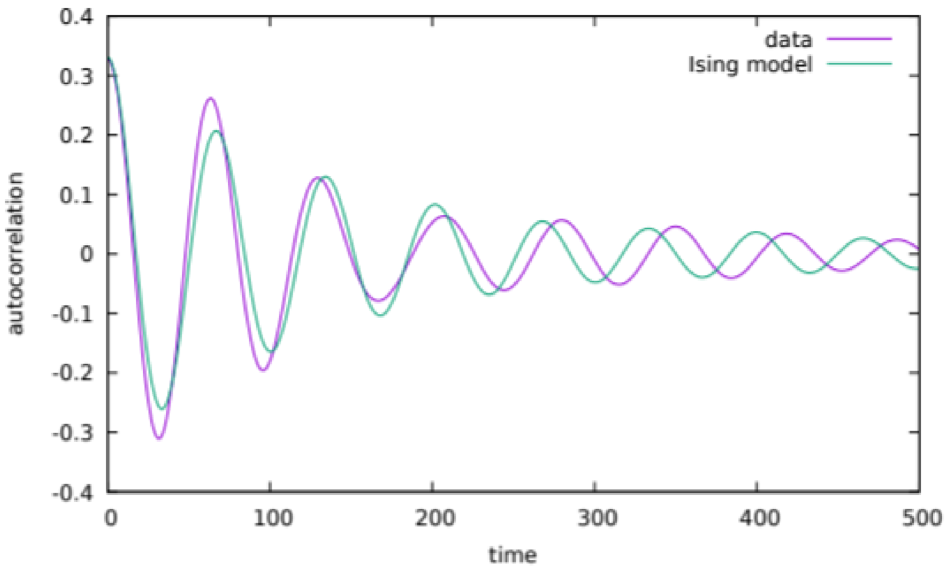


- Avalanche statistics (power-law exponents) somewhat similar to data
- Coexistence of avalanches and oscillations
- Robust to threshold and system size
- Small variation with feedback timescale and temperature

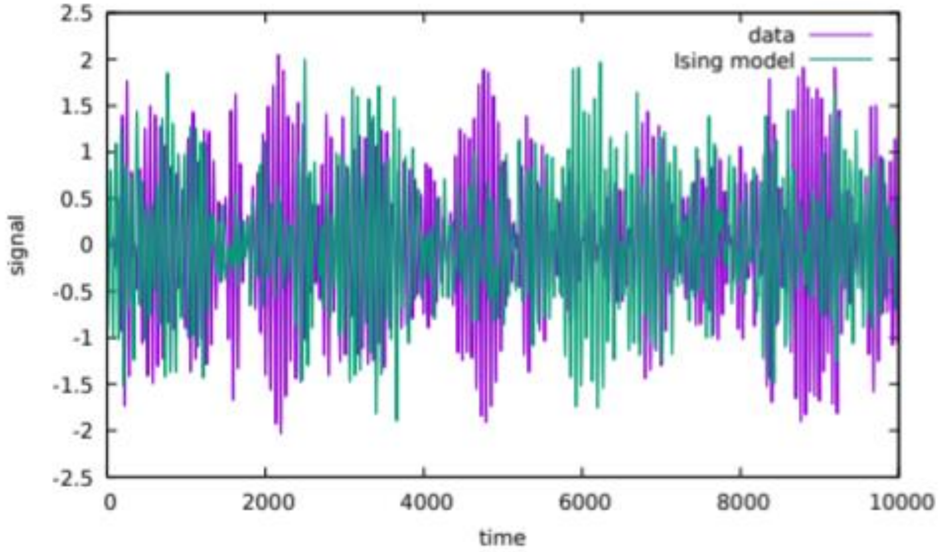
Comparison to MEG data

- **Data:** Human resting activity (eyes closed), 4 min broadband, 600 Hz sampling, 100 subjects, 273 gradiometers
- Fit model parameters $\boldsymbol{\tau}$ and $\boldsymbol{\beta}$ (feedback timescale, temp.) by matching autocorrelation functions
- Best fit for $\boldsymbol{\beta}$ **= 0.98-0.99**, a **slightly subcritical** state.

Signal autocorrelation function



Simulated / true signal



Conclusions and further directions

- A simple, analytically-tractable **negative feedback Ising** model shows **coexistence of oscillations and avalanches** near its critical point
- **Avalanche statistics** (power-law exponents) somewhat match data from neuronal networks
- **Feedback timescale and temperature** have the strongest influence on power-law exponents
- **MEG data** can be fitted to our model

- Effects of more local application of negative feedback/inhibition and different (realistic) network topology?

Thanks ☺

Questions?